

ASSIGNMENT ON CSE-422

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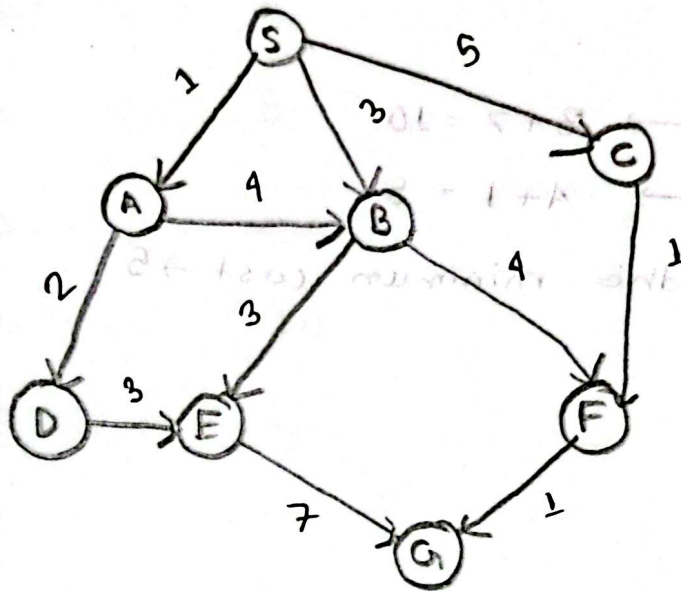
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Section: 11

Assignment No: 1

Semester : Summer 25

Answer to the Question No. 1



Heuristic values for states such that it is admissible.

The condition for admissibility be

$$h(n) \leq h^*(n)$$

States

→

$$F \rightarrow G$$

$$h^*(G) = 0, h^*(F) = 1$$

$$E \rightarrow G$$

$$h^*(E) = 7$$

$$C \rightarrow G$$

$$h^*(C) = 1 + 1 = 2$$

$$B \rightarrow G$$

$$h^*(B) \rightarrow 3 + 7 = 10$$

$$\rightarrow 4 + 1 = 5$$

we take the minimum cost $\rightarrow 5$

$$h^*(B) = 5$$

$$D \rightarrow G$$

$$h^*(D) = 10$$

$$A \rightarrow G$$

$$h^*(A) \rightarrow 12, 9, 14$$

$$\rightarrow 9$$

$$S \rightarrow G$$

$$h^*(S) \rightarrow 10, 8, 7$$

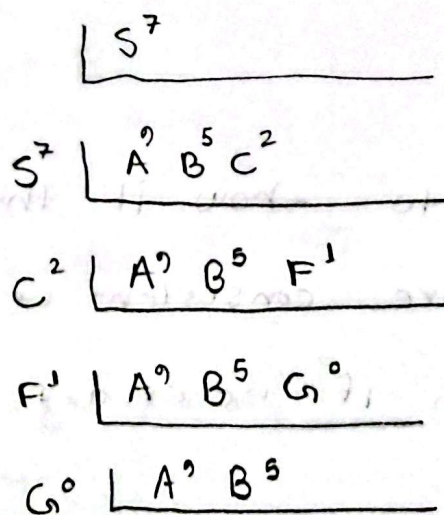
$$\rightarrow 7$$

So, After filling up the table with heuristic table we get,

<u>States</u>	<u>h(n)</u>
S	7
A	9
B	5
C	2
D	10
E	7
F	1
G	0

(B)

We are requested to perform Best First Search and A* on the following graph



Path ::

$S^7 \rightarrow C^2 \rightarrow F^1 \rightarrow G^0$

Now - we apply A^* on the following graph

$$\begin{array}{l}
 S^7 \mid S^{7+0} \\
 S^7 \mid A^{7+1} \quad B^{5+3} \quad C^{2+5} \\
 S^7 \mid A^{10} \quad B^8 \quad C^7 \\
 C^7 \mid A^{10} \quad B^8 \quad F^{1+6} \\
 F^7 \mid A^{10} \quad B^8 \quad G^{0+7} \\
 F^7 \mid A^{10} \quad B^8 \quad G^7 \\
 G^7 \mid A^{10} \quad B^8
 \end{array}$$

So, the path,

$$S^7 \rightarrow C^7 \rightarrow F^7 \rightarrow G^7$$

(C)

We are required to show if the heuristic values are consistent or not and correct them if necessary.

For consistency we know,

$$h(n) \leq c(n, n') + h(n')$$

<u>S</u>	<u>$h(n)$</u>	<u>$c(n, n')$</u>	<u>$h(n')$</u>	<u>Consistency</u>
$S \rightarrow A$	7	1	9	$7 \leq 10$
$S \rightarrow B$	7	3	5	$7 \leq 8$
$S \rightarrow C$	7	5	2	$7 \leq 7$
<u>A</u>				
$A \rightarrow B$	9	4	5	$9 \leq 9$
$A \rightarrow D$	9	2	10	$9 \leq 12$
<u>B</u>				
$B \rightarrow E$	5	3	7	$5 \leq 10$
$B \rightarrow F$	5	4	1	$5 \leq 5$
<u>C</u>				
$C \rightarrow F$	2	1	1	$2 \leq 2$
<u>D</u>				
$D \rightarrow E$	10	3	7	$10 \leq 10$
<u>F</u>				
$E \rightarrow G$	7	7	0	$7 \leq 7$
<u>F</u>				
$F \rightarrow G$	1	1	0	$1 \leq 1$

From the above discussions we can conclude that all values are consistent

(b)

(When a state is unreachable from the goal node.

(i) Heuristic Value $h(n) \rightarrow$ we know heuristic value should be admissible and thus will be,

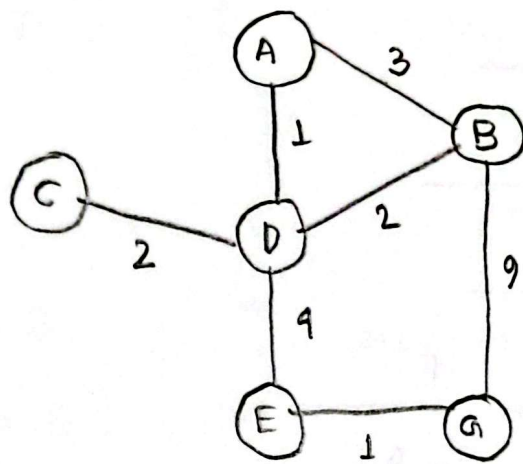
$$h(n) \leq h^*(n)$$

So, if we take a look at actual cost path we notice as it doesn't have a path it is $\rightarrow \infty$

$$\text{So, } \infty \leq \infty$$

Thus heuristic value should be 0 and the actual path cost should also be.

Answer to the Question No.2



(A)

Comparing heuristics

$h_1(n)$ $h_2(n)$

$$4 \leq 5$$

$$6 \leq 6$$

$$4 \leq 5$$

$$3 \leq 4$$

$$1 \leq 1$$

$$0 \leq 0$$

So, all the values of $h_2(n)$ is greater or equal to $h_1(n)$. So, $h_2(n)$ is dominant.
So, we take $h_2(n)$

(B)

We perform A* search with $h_2(n)$

$$\underline{A^{5+0}}$$

$$A^5 \mid \underline{D^5 \ B^9}$$

$$A^5 \mid \underline{D^5 \ B^9}$$

$$D^5 \mid \underline{B^9 \ C^{5+3} \ E^{1+5}}$$

$$D^5 \mid \underline{B^9 \ C^8 \ E^6}$$

$$E^6 \mid \underline{B^9 \ C^8 \ G^{6+0} \ (1)}$$

$$E^6 \mid \underline{B^9 \ C^8 \ G^6} \quad \text{Path}$$

$$G^6 \mid \underline{B^9 \ C^8 \ G^6}$$

$$A^5 \rightarrow D^5 \rightarrow E^6 \rightarrow G^6$$

(C)

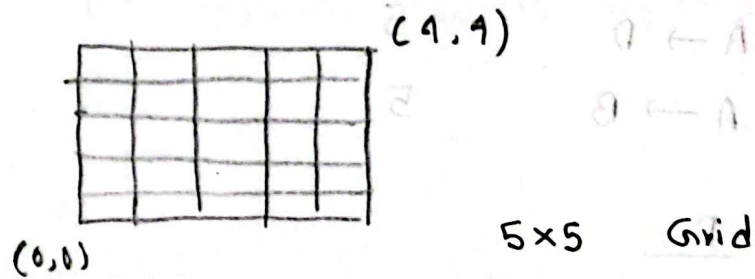
Now we are required to demonstrate whether the heuristic values are consistent or not. For consistency

$$h(n) \leq c(n, n') + h(n')$$

<u>A</u>	<u>$h(n)$</u>	<u>$c(n, n')$</u>	<u>$h(n')$</u>	<u>Consistency</u>
$A \rightarrow D$	5	1	4	$5 \leq 5$
$A \rightarrow B$	5	3	6	$5 \leq 9$
<u>D</u>				
$D \rightarrow C$	4	2	5	$4 \leq 7$
$D \rightarrow B$	4	2	6	$4 \leq 8$
$D \rightarrow E$	4	4	1	$4 \leq 5$
<u>B</u>				
$B \rightarrow G$	6	9	0	$6 \leq 9$
$B \rightarrow D$	6	2	4	$6 \leq 6$
$B \rightarrow A$	6	3	5	$6 \leq 8$
<u>C</u>				
$C \rightarrow D$	5	2	4	$5 \leq 6$
<u>E</u>				
$E \rightarrow G$	1	1	0	$1 \leq 1$
$E \rightarrow D$	1	4	4	$1 \leq 8$

So, we can see all the nodes are consistent.

Answer to the Question No. 3



horizontal $\rightarrow 1$

vertical $\rightarrow 1$

Diagonal $\rightarrow \sqrt{2}$

$h_1 \rightarrow$ Manhattan distance

$$|x_1 - x_2| + |y_1 - y_2|$$

$h_2 \rightarrow$ Euclidean Distance

$$\begin{aligned} & \sqrt{(x_2)^2 - x_1^2} + \sqrt{(y_2)^2 - y_1^2} \\ &= \sqrt{(x_2)^2 - (x_1)^2} + \sqrt{(y_2)^2 - (y_1)^2} \\ &= \sqrt{(x_2)^2 - x_1^2 + (y_2)^2 - y_1^2} \end{aligned}$$

(a)

We are to find out if the heuristics are admissible or not.

We use (1,1) and (2,2) as the initial and final point.

The actual cost to go from (1,1)

to (2,2) is 1 grid diagonally $= \sqrt{2}$

For h_1 ,

$$|1-2| + |1-2|$$

$$= 1+1 = 2$$

So, Actual cost $\leq h_1$

For h_2 ,

$$\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

$$= \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{2}$$

So, Actual cost and $h_2 \rightarrow h_2 \leq \text{Actual Cost}$

So, $h_2(n)$ is admissible

(b)

From the above question, we have come to the conclusion that $h_1(n)$ is not admissible and $h_2(n)$ is admissible. We run A^* using both heuristics.

For any node, Manhattan distance value is greater or equal to Euclidean.

While comparing as both cover the same distance, the one with the higher value / heuristic will cause in less nodes rather than the one with lower heuristic.

So, $h_1(n)$ will expand fewer nodes even if its not admissible

(C)

As $h_1(n)$ overestimates the true cost, multiplying it with 1.2 will further increase its value. So, no it

will not be optimal, and still remain in admissible.

Answer to the Question No.4

(a)

When the path is not relevant we tend to use Local Search.

The key properties of local search is :-

- (i) Key for Dynamic Problems
- (ii) Optimization Tasks which looks for improvement
- (iii) Search spaces where global search is difficult.
- (iv) Memory efficient.

(B)

Some examples of Local search is :-

- (1) Travelling Salesman
- (2) Layout Design
- (3) Knapsack Problem
- (4) Protein Folding

(C)

The difficulties we face during
of Hill Climb Search are:-

- 1) Local Maxima: The algorithm often gets fixed on a "peak" from where all neighbouring states are worse, but it is not the solution which is global maxima.
- (2) Plateaus: There is a "flat" area in the state space where all neighbouring states have the same value. Thus it gets stuck in a loop
- (3) Ridge: The search often gets stuck on a ridge. Thus algorithm cannot progress as it is limited

The remedies are also discussed below:-

- (1) To deal with local maxima, we operate the hill climbing algo

multiple times from different starting position. The best solution is then considered.

(2) To escape plateaus we can choose a random neighbour instead of the best one.

(3) We can also accept movement across neighbouring states to avoid ridges

(D)

We are required to demonstrate the darwicks Local Maxima and Plateau Hill Climbing Approach using 8-puzzle

Let's say h = number of correctly places tiles, so, goal state = 8

1	2	3
4	5	6
7	8	-

$h = 8$

So, for local maxima

→ not goal but any move → lower score from current state.

2	1	3
4	5	6
7	8	-

$h = 7$

(1 ↔ 2)

For Plateau

The movement doesn't help as it leads to the exact heuristic score :-

1	2	3
4	5	6
8	7	-

$h = 6$

1	2	3
4	5	6
8	-	7

$h = 6$



So, h still remains $h = 6$

(E)

Simulated Annealing is an important technique in finding solution to optimization problems. The key steps are:-

(1) Initialization

→ Random Initial state and high temperature

(2) Loop / Cooling Rate

(3) Termination or end temperature

(F)

The use case of probability in Simulated Annealing is that it determines whether to accept a worse situation than the current one. It is determined using Metropolis Hastings Algorithm.

$$P(\text{accept } S') = \min(1, e^{-\frac{\Delta E}{T}})$$

S' = new state

S = current state

Unlike greedy algorithm, Simulated

Annealing allows for occasional

high energy to escape local optima

So, if new solution better $\rightarrow$ (low energy) $\rightarrow$ always accepted

If new solution worse (higher energy)

$\rightarrow$ accepted with certain probability

$$e^{-\Delta E / T}$$

(G)

The relationship between Temperature (T) and probabilistic value is exponential and inverse

$$P = e^{-\Delta E / T}$$

$$\Delta E = E_{\text{new}} - E_{\text{current}}$$

T = Temperature

When T is high $\rightarrow$ probability of accepting a worse decision high. Temperature is low, the probability of accepting worse is also low